

Project Name:	Dual Ocular Disease Screener
Team:	EyeQ
Project Description:	For primary care physicians, optometrists, and patients who may be at risk and need a quick way to know if they should see an eye specialist, the Dual Ocular Disease Screener is a web application that analyzes a fundus image of the eye and gives separate risk results for cataract and glaucoma. Our system uses one shared deep learning model to analyze the image for both eye conditions. It will also provide a confidence level and a heatmap to show which parts of the image were important for the result. If the image quality is not good or the model is not confident about its prediction, the system will recommend referring the patient to a specialist instead of giving a confident result. Different from models that are mainly tested on the same type of data used during development, our project will also test the model on independent external datasets. This will help us understand how well the model works on new data and where its performance may decrease.
Benefit Outcomes:	• Screen for cataract and glaucoma from one fundus image. • Help clinicians make faster referral decisions. • Reduce confident mistakes by referring to uncertain or low-quality images. • Use heatmaps to help users understand which areas of the image affected the result. • Test the model on external datasets to better understand how reliable it is on new data.
Github Link:	https://github.com/htmw/F2026-Team1/wiki